

# Measurement

## MCQ

1. Measurement is the process of

- a) Comparing objects
- b) Associating numbers with physical quantities
- c) Observing phenomena
- d) Estimating values

Answer: b) Associating numbers with physical quantities

2. Measurement is fundamental to

- a) Arts only
- b) Sciences and engineering
- c) Literature
- d) Music

Answer: b) Sciences and engineering

3. Measurement gives a quantity its

- a) Shape
- b) Colour
- c) Numerical value and unit
- d) Texture

Answer: c) Numerical value and unit

4. The value obtained on measuring a quantity is called its

- a) Unit
- b) Dimension
- c) Magnitude
- d) Density

Answer: c) Magnitude

5. Units are used to

- a) Increase accuracy only
- b) Describe what numbers represent
- c) Change values
- d) Remove errors

Answer: b) Describe what numbers represent

6. Names of SI units usually start with

- a) Capital letters
- b) Uppercase letters
- c) Lowercase letters
- d) Greek letters

Answer: c) Lowercase letters

7. In “degree Celsius”, which word is capitalized?

- a) Degree
- b) Celsius
- c) Both
- d) None

Answer: b) Celsius

8. Units are important mainly to

- a) Save time
- b) Measure and communicate quantities
- c) Increase size
- d) Reduce cost

Answer: b) Measure and communicate quantities

9. A physical quantity is a property that can be

- a) Seen only
- b) Touched only

c) Measured and expressed numerically

d) Imagined

Answer: c) Measured and expressed numerically

10. Physical quantities are classified into

a) Scalar and vector

b) Fundamental and derived

c) Large and small

d) Simple and complex

Answer: b) Fundamental and derived

11. Fundamental quantities are also called

a) Derived quantities

b) Base quantities

c) Secondary quantities

d) Complex quantities

Answer: b) Base quantities

12. Number of fundamental quantities in SI system is

a) 5

b) 6

c) 7

d) 8

Answer: c) 7

13. Units of fundamental quantities are called

a) Derived units

b) Standard units

c) Base units

d) Mixed units

Answer: c) Base units

14. Derived quantities depend on

- a) Time only
- b) Mass only
- c) Fundamental quantities
- d) Temperature only

Answer: c) Fundamental quantities

15. Area is a

- a) Fundamental quantity
- b) Scalar quantity
- c) Derived quantity
- d) Vector quantity

Answer: c) Derived quantity

16. Area is calculated using

- a) Length  $\times$  length
- b) Mass  $\times$  volume
- c) Speed  $\times$  time
- d) Force  $\times$  distance

Answer: a) Length  $\times$  length

17. SI unit of area is

- a) m
- b)  $m^3$
- c)  $m^2$
- d) cm

Answer: c)  $m^2$

18. Objects with definite shape and size are called

- a) Irregular objects
- b) Regular objects

c) Random objects

d) Mixed objects

Answer: b) Regular objects

19. Which is a regular shape?

a) Leaf

b) Rock

c) Rectangle

d) Cloud

Answer: c) Rectangle

20. Area of irregular shapes can be found by

a) Formula only

b) Guessing

c) Dividing into regular shapes

d) Weighing

Answer: c) Dividing into regular shapes

21. Graphical method is mainly used to find area of

a) Regular figures

b) Circular figures

c) Irregular figures

d) Spherical figures

Answer: c) Irregular figures

22. In graphical method, complete squares are taken as

a) A

b) B

c) C

d) D

Answer: a) A

23. Volume is the amount of

- a) Mass in an object
- b) Space occupied by an object
- c) Weight of an object
- d) Density of an object

Answer: b) Space occupied by an object

24. Volume is a

- a) Fundamental quantity
- b) Derived quantity
- c) Base quantity
- d) Independent quantity

Answer: b) Derived quantity

25. SI unit of volume is

- a)  $m^2$
- b)  $m^3$
- c) litre
- d) cm

Answer: b)  $m^3$

26. A cube and a cylinder differ in volume because of

- a) Colour
- b) Shape
- c) Material
- d) Weight

Answer: b) Shape

27. Liquids do not have

- a) Volume
- b) Mass

c) Definite shape

d) Density

Answer: c) Definite shape

28. Volume of liquids is measured using

a) Ruler

b) Balance

c) Measuring cylinder

d) Thermometer

Answer: c) Measuring cylinder

29. Capacity of a container means

a) Its mass

b) Its weight

c) Maximum volume it can hold

d) Its density

Answer: c) Maximum volume it can hold

30. Unit commonly used for measuring liquids is

a) Kilogram

b) Metre

c) Litre

d) Newton

Answer: c) Litre

31. One litre is equal to

a) 10 cc

b) 100 cc

c) 1000 cc

d) 10,000 cc

Answer: c) 1000 cc

32.1 ml is equal to

- a)  $1 \text{ m}^3$
- b)  $1 \text{ cm}^3$
- c)  $10 \text{ cm}^3$
- d)  $0.1 \text{ cm}^3$

Answer: b)  $1 \text{ cm}^3$

33. Irregular solid volumes are measured by

- a) Graphical method
- b) Weighing method
- c) Water displacement method
- d) Heating method

Answer: c) Water displacement method

34. Water displacement method was discovered by

- a) Newton
- b) Einstein
- c) Archimedes
- d) Galileo

Answer: c) Archimedes

35. Initial water level in a cylinder is called

- a) V
- b)  $V_2$
- c)  $V_1$
- d) D

Answer: c)  $V_1$

36. Final water level after immersing an object is

- a)  $V_1$
- b)  $V_2$

c) A

d) C

Answer: b)  $V_2$

37. Volume of an irregular solid is equal to

a)  $V_1 + V_2$

b)  $V_1 \times V_2$

c)  $V_2 - V_1$

d)  $V_1 \div V_2$

Answer: c)  $V_2 - V_1$

38. Density is defined as

a) Mass  $\times$  volume

b) Volume  $\div$  mass

c) Mass  $\div$  volume

d) Mass + volume

Answer: c) Mass  $\div$  volume

39. SI unit of density is

a)  $\text{g/cm}^3$

b)  $\text{kg/m}^2$

c)  $\text{kg/m}^3$

d)  $\text{g/m}^3$

Answer: c)  $\text{kg/m}^3$

40. CGS unit of density is

a)  $\text{kg/m}^3$

b)  $\text{g/cm}^3$

c)  $\text{g/m}^2$

d)  $\text{kg/cm}^3$

Answer: b)  $\text{g/cm}^3$

41. A material with higher density is called

- a) Rarer
- b) Lighter
- c) Denser
- d) Softer

Answer: c) Denser

42. Air is considered

- a) Denser than water
- b) Rarer than water
- c) Equal to water
- d) Solid

Answer: b) Rarer than water

43. Objects float on water if their density is

- a) Greater than water
- b) Equal to water
- c) Less than water
- d) Zero

Answer: c) Less than water

44. Oil floats on water because oil is

- a) Heavier
- b) Denser
- c) Less dense
- d) Solid

Answer: c) Less dense

45. Density depends on

- a) Shape only
- b) Colour only

c) Mass and volume

d) Temperature only

Answer: c) Mass and volume

46. Mass of an object can be found if

a) Volume and density are known

b) Shape is known

c) Colour is known

d) Area is known

Answer: a) Volume and density are known

47. Distances of stars are measured in

a) Metres

b) Kilometres

c) Light years

d) Centimetres

Answer: c) Light years

48. Astronomical unit is used to measure distances within

a) Earth

b) Galaxy

c) Solar system

d) Universe

Answer: c) Solar system

49. One astronomical unit is the mean distance between

a) Earth and Moon

b) Earth and Sun

c) Sun and Mars

d) Moon and Sun

Answer: b) Earth and Sun

50. Value of one AU is approximately

- a) 14.96 million km
- b) 149.6 million km
- c) 1.496 million km
- d) 1496 million km

Answer: b) 149.6 million km

51. Perihelion is the position where Earth is

- a) Closest to the Sun
- b) Farthest from the Sun
- c) Closest to the Moon
- d) Farthest from stars

Answer: a) Closest to the Sun

52. Aphelion is the position where Earth is

- a) Closest to Sun
- b) Closest to Moon
- c) Farthest from Sun
- d) Inside Sun

Answer: c) Farthest from Sun

53. Light year is a unit of

- a) Time
- b) Speed
- c) Distance
- d) Energy

Answer: c) Distance

54. Speed of light in vacuum is

- a)  $3 \times 10^6$  m/s
- b)  $3 \times 10^7$  m/s

c)  $3 \times 10^8$  m/s

d)  $3 \times 10^9$  m/s

Answer: c)  $3 \times 10^8$  m/s

55. One light year is the distance travelled by light in

a) One day

b) One month

c) One year

d) One hour

Answer: c) One year

56. Total seconds in one non-leap year is approximately

a)  $3.153 \times 10^6$

b)  $3.153 \times 10^7$

c)  $3.153 \times 10^8$

d)  $3.153 \times 10^9$

Answer: b)  $3.153 \times 10^7$

57. Value of one light year in metres is

a)  $9.46 \times 10^{12}$

b)  $9.46 \times 10^{13}$

c)  $9.46 \times 10^{15}$

d)  $9.46 \times 10^{17}$

Answer: c)  $9.46 \times 10^{15}$

58. Proxima Centauri is

a) A planet

b) A galaxy

c) The nearest star

d) A comet

Answer: c) The nearest star

59.Distance of Proxima Centauri from Earth is about

- a) 1 light year
- b) 2 light years
- c) 4.23 light years
- d) 10 light years

Answer: c) 4.23 light years

60.Alpha Centauri is approximately

- a) 1 light year away
- b) 4.3 light years away
- c) 10 light years away
- d) 25,000 light years away

Answer: b) 4.3 light years away

61.Distance of Earth from galactic centre is about

- a) 2,500 light years
- b) 10,000 light years
- c) 25,000 light years
- d) 50,000 light years

Answer: c) 25,000 light years

62.Parsec is used to measure

- a) Liquid volume
- b) Atomic distance
- c) Astronomical distances
- d) Length of objects

Answer: c) Astronomical distances

63.Volume of a cube depends on

- a) One side only
- b) Length and breadth

c) Side  $\times$  side  $\times$  side

d) Radius and height

Answer: c) Side  $\times$  side  $\times$  side

64. Volume of a cylinder depends on

a) Radius and height

b) Diameter only

c) Side only

d) Area only

Answer: a) Radius and height

65. Measuring cylinder readings are usually marked in

a) kg

b) cm

c) ml

d) m<sup>2</sup>

Answer: c) ml

66. Density triangle shows relationship between

a) Area, volume, mass

b) Mass, volume, density

c) Speed, distance, time

d) Force, mass, acceleration

Answer: b) Mass, volume, density

67. If mass increases and volume remains constant, density

a) Decreases

b) Remains same

c) Increases

d) Becomes zero

Answer: c) Increases

68.If volume increases and mass remains constant, density

- a) Increases
- b) Decreases
- c) Remains same
- d) Doubles

Answer: b) Decreases

69.Density helps to identify

- a) Shape of object
- b) Colour of object
- c) Material of object
- d) Size of object

Answer: c) Material of object

70.Water is denser than

- a) Iron
- b) Oil
- c) Copper
- d) Aluminium

Answer: b) Oil

71.Volume of liquid equals

- a) Mass of container
- b) Capacity used
- c) Weight of liquid
- d) Density of liquid

Answer: b) Capacity used

72.Measuring long space distances in km is difficult because they are

- a) Small
- b) Moderate

c) Extremely large

d) Equal

Answer: c) Extremely large

73. AU is not suitable for measuring distances between

a) Sun and Earth

b) Planets

c) Stars and galaxies

d) Earth and Mars

Answer: c) Stars and galaxies

74. Density of a substance remains constant if

a) Mass and volume change proportionally

b) Only mass changes

c) Only volume changes

d) Shape changes

Answer: a) Mass and volume change proportionally

75. Which quantity is not derived?

a) Area

b) Volume

c) Density

d) Length

Answer: d) Length

76. Length is measured in

a)  $\text{m}^3$

b)  $\text{m}^2$

c) metre

d) litre

Answer: c) metre

77. Area of a rectangle is

- a) Length + breadth
- b) Length  $\times$  breadth
- c)  $2(\text{length} + \text{breadth})$
- d) Length  $\div$  breadth

Answer: b) Length  $\times$  breadth

78. A measuring beaker is mainly used to measure

- a) Mass
- b) Temperature
- c) Liquid volume
- d) Length

Answer: c) Liquid volume

79. Volume is also called

- a) Weight
- b) Area
- c) Capacity
- d) Density

Answer: c) Capacity

80. Measurement helps to make observations

- a) Confusing
- b) Uniform and meaningful
- c) Unclear
- d) Irrelevant

Answer: b) Uniform and meaningful